

```
In [1]: import numpy as np
import pandas as pd
```

```
In [5]: df = pd.read_csv('student_clustering.csv')
print("The shape of data is",df.shape)
df.head()
```

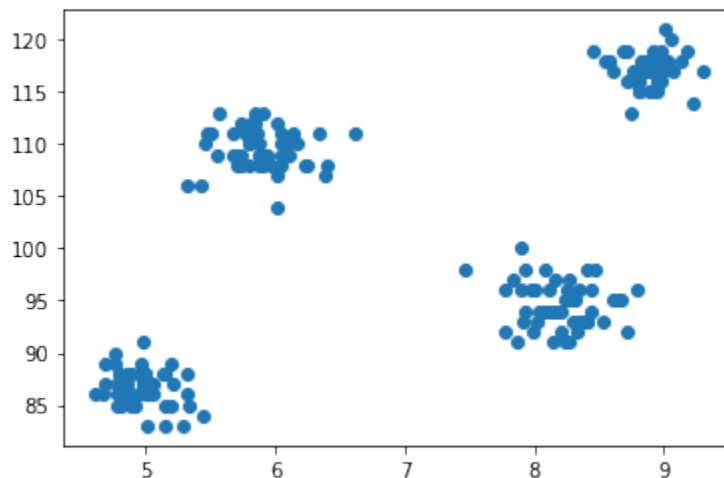
The shape of data is (200, 2)

```
Out[5]:
```

	cgpa	iq
0	5.13	88
1	5.90	113
2	8.36	93
3	8.27	97
4	5.45	110

```
In [6]: import matplotlib.pyplot as plt
plt.scatter(df['cgpa'],df['iq'])
```

Out[6]: <matplotlib.collections.PathCollection at 0x193e8308640>



```
In [7]: from sklearn.cluster import KMeans
```

```
In [8]: wcss = []

for i in range(1,11):
    km = KMeans(n_clusters=i)
    km.fit_predict(df)
    wcss.append(km.inertia_)
```

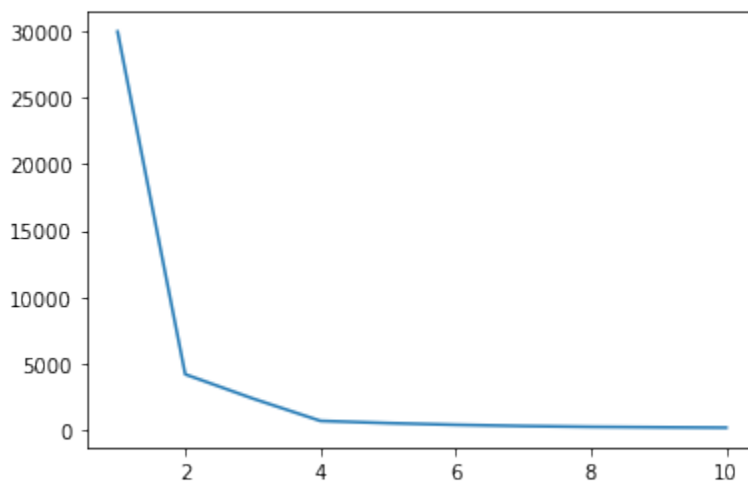
C:\Users\91842\anaconda3\lib\site-packages\sklearn\cluster_kmeans.py:881: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=1.
warnings.warn(

```
In [9]: wcss
```

```
Out[9]: [29957.898287999997,  
         4184.141269999999,  
         2362.7133490000015,  
         681.9696600000001,  
         514.1616803171117,  
         388.8524026875981,  
         295.4391895943191,  
         234.48687343199217,  
         199.99120032567834,  
         171.40590752168507]
```

```
In [11]: plt.plot(range(1,11),wcss)
```

```
Out[11]: [matplotlib.lines.Line2D at 0x193eba778e0>]
```



```
In [13]: X = df.iloc[:,:].values  
km = KMeans(n_clusters=4)  
y_means = km.fit_predict(X)
```

```
In [14]: y_means
```

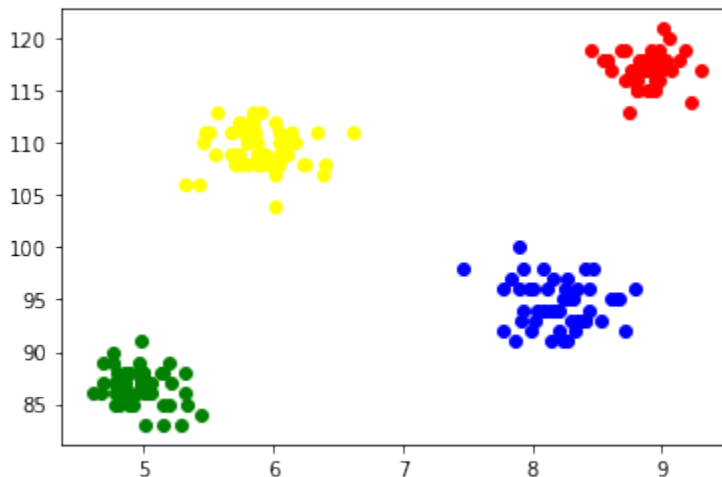
```
Out[14]: array([2, 3, 0, 0, 3, 3, 0, 1, 3, 0, 2, 3, 0, 2, 3, 0, 3, 0, 3, 3, 0, 2,  
                0, 2, 2, 0, 2, 1, 0, 3, 1, 3, 1, 3, 0, 0, 1, 3, 2, 3, 2, 0, 0, 2,  
                1, 1, 0, 3, 1, 3, 2, 2, 1, 0, 1, 3, 3, 1, 3, 1, 3, 0, 0, 1, 2, 1,  
                0, 2, 3, 0, 3, 1, 0, 2, 3, 1, 3, 1, 2, 0, 0, 1, 3, 2, 1, 2, 1, 3,  
                1, 3, 1, 1, 0, 2, 0, 0, 1, 0, 2, 1, 3, 2, 2, 1, 2, 2, 0, 2, 1, 1,  
                0, 1, 3, 3, 0, 1, 0, 3, 1, 2, 2, 3, 0, 1, 0, 2, 0, 3, 2, 0, 0, 3,  
                2, 2, 3, 1, 3, 2, 0, 0, 0, 2, 3, 2, 2, 1, 2, 1, 3, 2, 1, 2, 1, 1,  
                2, 0, 3, 1, 3, 0, 2, 1, 3, 0, 1, 2, 3, 2, 2, 1, 1, 3, 1, 2, 2, 0,  
                1, 3, 2, 1, 1, 3, 3, 3, 0, 2, 0, 0, 1, 3, 0, 0, 2, 2, 0, 2, 1, 3,  
                3, 1])
```

```
In [22]: X[y_means == 3,1]
```

```
Out[22]: array([[113., 110., 109., 110., 110., 108., 111., 111., 113., 106., 112.,
                106., 108., 113., 108., 109., 108., 111., 109., 104., 111., 109.,
                109., 112., 111., 109., 108., 110., 109., 111., 108., 109., 109.,
                111., 112., 109., 111., 107., 111., 110., 108., 109., 110., 108.,
                108., 110., 107., 111., 112., 108.]])
```

```
In [23]: plt.scatter(X[y_means == 0,0],X[y_means == 0,1],color='blue')
plt.scatter(X[y_means == 1,0],X[y_means == 1,1],color='red')
plt.scatter(X[y_means == 2,0],X[y_means == 2,1],color='green')
plt.scatter(X[y_means == 3,0],X[y_means == 3,1],color='yellow')
```

```
Out[23]: <matplotlib.collections.PathCollection at 0x193ec760fd0>
```



K-Means on 3-D Data

```
In [24]: from sklearn.datasets import make_blobs

centroids = [(-5,-5,5),(5,5,-5),(3.5,-2.5,4),(-2.5,2.5,-4)]
cluster_std = [1,1,1,1]

X,y = make_blobs(n_samples=200,cluster_std=cluster_std,centers=centroids,n_fea
```

```
In [25]: X
```

```
Out[25]: array([[ 4.33424548,  3.32580419, -4.17497018],
 [-3.32246719,  3.22171129, -4.625342  ],
 [-6.07296862, -4.13459237,  2.6984613 ],
 [ 6.90465871,  6.1110567 , -4.3409502 ],
 [-2.60839207,  2.95015551, -2.2346649 ],
 [ 5.88490881,  4.12271848, -5.86778722],
 [-4.68484061, -4.15383935,  4.14048406],
 [-1.82542929,  3.96089238, -3.4075272 ],
 [-5.34385368, -4.95640314,  4.37999916],
 [ 4.91549197,  4.70263812, -4.582698  ],
 [-3.80108212, -4.81484358,  4.62471505],
 [ 4.6735005 ,  3.65732421, -3.88561702],
 [-6.23005814, -4.4494625 ,  5.79280687],
 [-3.90232915,  2.95112294, -4.6949209 ],
 [ 3.72744124,  5.31354772, -4.49681519],
 [-3.3088472 ,  3.05743945, -3.81896126],
 [ 2.70273021, -2.21732429,  3.17390257],
 [ 4.06438286, -0.36217193,  3.214466  ],
 [ 4.69268607, -2.73794194,  5.15528789],
 [ 4.1210827 , -1.5438783 ,  3.29415949],
 [-6.61577235, -3.87858229,  5.40890054],
 [ 3.05777072, -2.17647265,  3.89000851],
 [-1.48617753,  0.27288737, -5.6993336 ],
 [-5.3224172 , -5.38405435,  6.13376944],
 [-5.26621851, -4.96738545,  3.62688268],
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 [ 5.12910158,  6.6169496 , -4.49725912],
 [-2.46505641,  3.95391758, -3.33831892],
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 [ 3.87798127,  4.48290554, -5.99702683],
 [ 4.10944442,  3.8808846 , -3.0439211 ],
 [-6.09989127, -5.17242821,  4.12214158],
 [-3.03223402,  3.6181334 , -3.3256039 ],
 [ 7.44936865,  4.45422583, -5.19883786],
 [-4.47053468, -4.86229879,  5.07782113],
 [-1.46701622,  2.27758597, -2.52983966],
 [ 3.0208429 , -2.14983284,  4.01716473],
 [ 3.82427424, -2.47813716,  3.53132618],
 [-5.74715829, -3.3075454 ,  5.05080775],
 [-1.51364782,  2.03384514, -2.61500866],
 [-4.80170028, -4.88099135,  4.32933771],
 [ 6.55880554,  5.1094027 , -6.2197444 ],
 [-1.48879294,  1.02343734, -4.14319575],
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```

[-3.03289825, -6.85798186, 6.23616403],
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[-3.25518824, -5.7612069 , 5.3190391],

```

[-4.95778625, -4.41718479, 3.89938082],
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[-4.09914405, -5.68372786, 4.87710977],
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[ 1.38583608, -2.91163916, 5.27852808],
[ 4.42001793, -2.69505734, 4.80539342],
[ 4.71269214, 5.68006984, -5.3198016 ],
[-4.13744959, 6.4586027 , -3.35135636],
[-5.20889423, -4.41337681, 5.83898341],
[ 2.6194224 , -2.77909772, 5.62284909],
[-1.3989998 , 3.28002714, -4.6294416 ]])

```

```

In [26]: import plotly.express as px
fig = px.scatter_3d(x=X[:,0], y=X[:,1], z=X[:,2])
fig.show()

```

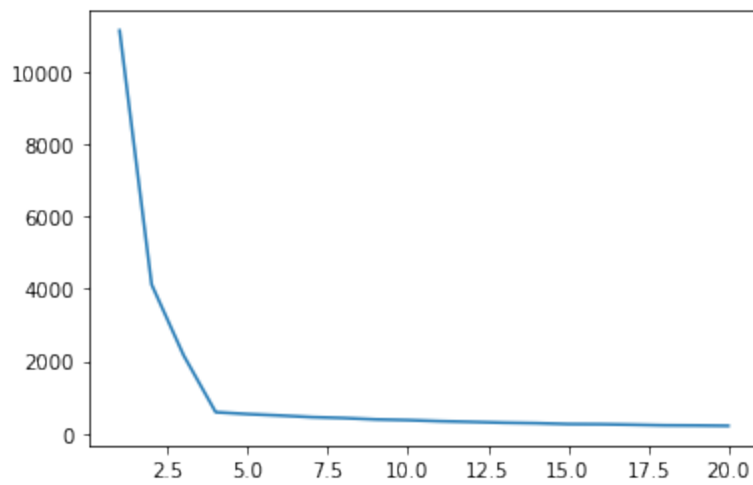
```
In [27]: wcss = []  
         for i in range(1,21):  
             km = KMeans(n_clusters=i)  
             km.fit_predict(X)  
             wcss.append(km.inertia_)
```

C:\Users\91842\anaconda3\lib\site-packages\sklearn\cluster_kmeans.py:881: User Warning:

KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=1.

```
In [28]: plt.plot(range(1,21),wcss)
```

```
Out[28]: [matplotlib.lines.Line2D at 0x193ef93f940>]
```

```
In [29]: km = KMeans(n_clusters=4)
y_pred = km.fit_predict(X)
```

```
In [30]: df = pd.DataFrame()

df['col1'] = X[:,0]
df['col2'] = X[:,1]
df['col3'] = X[:,2]
df['label'] = y_pred
```

```
In [31]: fig = px.scatter_3d(df,x='col1', y='col2', z='col3',color='label')
fig.show()
```

